

Tokhirzhon Tashmatov

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SUMMARY

AI/ML engineer focused on **LLM systems and their evaluation**. I ship production LLM applications end-to-end (agents, RAG, retrieval) and build **adversarial-evaluation and red-teaming tooling** that stress-tests them and measures failure modes. Strong Python and data-engineering foundation; bias toward building, testing, and measuring what actually works.

EXPERIENCE

Machine Learning / AI Engineer *ConeRed*

Jul. 2025 – Present
Remote (Spain)

- Built and open-sourced an **LLM red-teaming toolkit** that fires **75 adversarial prompt-injection attacks** at any HTTP chat endpoint with **layered detection** (regex + vector similarity + ML classifier + **LLM-as-judge**) and a reproducible verdict report.
- Architected a multi-service AI knowledge platform (FastAPI, Next.js/TS, PostgreSQL, Redis, Qdrant, Celery) as a containerized **docker-compose** stack with nginx TLS ingress; shipped features end-to-end without breaking production.
- Built **LangGraph** agent pipelines (context → summarize → intent → retrieval → response) over Voyage embeddings in Qdrant, serving multilingual real-time Q&A to **3,500+** users across web, Telegram, and an embeddable widget.
- Engineered an agentic backend for autonomous B2B research and outreach with tool-use over LinkedIn (ProxyCurl), Perplexity, and web scraping (JINA), auto-generating scored reports into Google Docs/Sheets.

Data Engineer *EPAM Systems*

Jun. 2024 – Dec. 2024
Bishkek, KG (Hybrid)

- Designed and maintained ETL pipelines processing **500 GB/day** of log data in Python + SQL on PostgreSQL.
- Cut query latency **22%** on high-load services by optimizing schemas, indexes, and rewriting queries from measured plans; built **FastAPI** data-retrieval and pipeline-monitoring tools for analyst-facing visibility.

Backend Developer *Premium Soft*

Jan. 2024 – May 2024
Remote (Sofia, BG)

- Developed **FastAPI** backend services for AI-powered internal products with clean architecture; implemented REST APIs for data ingestion, processing, and LLM inference pipelines integrated into existing systems.

Android Developer *Makers Incubator*

Jun. 2023 – Nov. 2023
Bishkek, KG

- Engineered an Android e-learning app (Kotlin, Firebase), lifting course registrations **30%**; led full-cycle development and cut delivery time **20%** vs prior projects.

PROJECTS & OPEN SOURCE

GRANAT — Tech Lead, choir-synthesis & vocal-alignment audio pipeline

Python, DSP/ML

- Lead the project end-to-end for a well-known recording artist's music project: a pipeline that aligns and mixes phone-recorded solo vocal takes from many singers into a coherent studio track (onset-envelope DTW + WSOLA time-stretch, ECAPA-TDNN speaker embeddings, Whisper alignment); own architecture, roadmap, and delivery.

Aegis Security / injection-shield — github.com/Tokhirzhon/Aegis-Security

Python, FastAPI

- Open-source AI-security platform; its **injection-shield** module runs 75 adversarial prompt-injection attacks with stacked detection (regex + embeddings + ML classifier + LLM-as-judge) and reproducible reporting; packaged on PyPI (GPLv3) with a browser GUI.

GeoGuard — AI geospatial landslide/mudflow prediction — github.com/Tokhirzhon/GeoGuard

Python, ML

- Multi-modal predictor (XGBoost/RF/NB over 17 terrain features + CNNs on satellite imagery); tabular AUC > 0.94, EfficientNetB0 85.4%. Open-sourced under Apache-2.0.

CoVibeCode (contributor) — local-first desktop app for AI coding agents

Tauri v2, Svelte 5, Rust

- Contribute to a cross-platform desktop client that runs and manages AI coding agents (Claude Code, Codex) locally.

Randevu (solo founder) — intent-based social matching, Telegram Mini App

Next.js, FastAPI

- Built end-to-end: 15-question profiling, embedding + LLM matching with “why you match” explanations; native iOS/Android in progress.

RESEARCH

Bachelor’s Thesis: Landslide Susceptibility Prediction for Southern Kyrgyzstan

2025

- Multi-modal predictor of mountain natural hazards; published on ResearchGate. Tabular ensemble AUC > 0.94; EfficientNetB0 85.4% on satellite imagery alone.

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript, SQL, Kotlin, C/C++
LLM / AI: RAG, GraphRAG, LangGraph agents, tool-calling, structured outputs, prompt engineering, **LLM-as-judge** evals, prompt-injection defense; OpenAI, Anthropic, Gemini, OpenRouter, Voyage, JINA
Backend: FastAPI, Flask, Pydantic, SQLAlchemy, Alembic, Celery, APScheduler
Data & Infra: PostgreSQL (indexing, query plans), Redis, ETL pipelines, Qdrant, Pinecone, FalkorDB; Docker, docker-compose, GitHub Actions, nginx, AWS, GCP
Frontend: React, Next.js, Tailwind (familiar: Vue 3 + Quasar)
Tools: Claude Code, Cursor, Copilot daily
Spoken: English (C1), Russian (fluent), Kyrgyz (native), Uzbek (native), Tajik (basic)

EDUCATION

American University of Central Asia
B.A. in Software Engineering — full-ride U.S.-CAEF scholarship (top 1% of applicants)

Aug. 2021 – Jun. 2025
Bishkek, KG

American University in Bulgaria
Exchange semester, Computer Science — OSUN Mobility Scholarship

Jan. 2024 – May 2024
Blagoevgrad, BG

HONORS

Winner, Cyber-Security Hackathon — AUCA, ITF (Slovenia), U.S. Embassy Bishkek · **FLEX finalist** (U.S. Dept. of State) · **1st place, National Technology Creation Olympiad** (2020)